

CS-23334 FUNDAMENTALS OF DATA SCIENCE

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Experiment 8

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Experiment to understand Logistic Regression for a given data set.

Aim:

To conduct an experiment to understand Logistic Regression for a given dataset

Description:

Understand Logistic Regression algorithm for a given dataset

Algorithm:

Step 1: Define the Classification Problem and Select Features

Step 2: Split the Dataset into Training and Testing Sets

Step 3: Train the Logistic Regression Model

Step 4: Predict and Evaluate Model Performance

Step 5: Interpret Coefficients and Visualize Results

About Dataset:

This dataset represents user information with attributes including User ID, Gender, Age, Estimated Salary, and a binary Purchased indicator showing whether the user made a purchase (1) or not (0).

Code With Output:

```
import numpy as np
import pandas as pd
df=pd.read_csv(r'D:\REC 2nd Year\Data Science\Data Sets\
Social_Network_Ads.csv')
df
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

[400 rows x 5 columns]

```
df.head()
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```
features=df.iloc[:,[2,3]].values
label=df.iloc[:,4].values
features[:10]
```

```
array([[ 19, 19000],
       [ 35, 20000],
       [ 26, 43000],
       [ 27, 57000],
       [ 19, 76000],
       [ 27, 58000],
       [ 27, 84000],
       [ 32, 150000],
       [ 25, 33000],
       [ 35, 65000]])
```

```

label
array([0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1,
1,
      1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,
      0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0,
0,
      0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0,
0,
      0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,
0,
      0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,
      0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0,
0,
      0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0,
0,
      0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0,
      0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0,
1,
      0, 1, 1, 1, 0, 0, 1, 1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 0, 0, 1, 1, 1,
0,
      1, 1, 0, 1, 0, 1, 0, 1, 0, 0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1, 1,
0,
      1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0, 1,
1,
      0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0,
1,
      1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 1, 0, 0, 0, 1, 1,
1,
      0, 1, 0, 0, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 1, 1,
0,
      1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0, 1,
1,
      0, 1, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1,
1,
      1, 1, 0, 1])

```

```

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

for i in range(1, 401):
    x_train, x_test, y_train, y_test = train_test_split(features,
label, test_size=0.2, random_state=i)
    model = LogisticRegression()
    model.fit(x_train, y_train)
    train_score = model.score(x_train, y_train)
    test_score = model.score(x_test, y_test)

    if test_score > train_score:
        print("Test {:.4f} Train {:.4f} Random State
{}".format(test_score, train_score, i))

Test 0.9000 Train0.8406 Random State 4
Test 0.8625 Train0.8500 Random State 5

```

Result:

Thus experiment was conducted to understand logistic regression for a dataset